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INNOVATION | TECHNOLOGY | BETTER TOMORROW

REGENERATIVE BRAKING SYSTEM

Recover Energy | Improve Efficiency | Drive a Greener Future



More Range
Less Energy Loss
Lower Emissions

OBJECTIVE

To recover the kinetic energy of the vehicle during braking and convert it into electrical energy, which is stored in the battery, improving overall energy efficiency and range.

KEY FEATURES

- Recovers braking energy
- Increases driving range
- Reduces brake wear
- Improves fuel/energy efficiency
- Works with electric & hybrid vehicles

APPLICATIONS

- Electric Vehicles (EVs)
- Hybrid Electric Vehicles (HEVs)
- Regenerative braking in trains
- Electric buses and commercial vehicles
- Industrial electric machinery

ADVANTAGES

- Higher energy efficiency
- Extended battery life
- Less heat generation
- Reduced emissions
- Lower maintenance cost

1. INTRODUCTION & WORKING PRINCIPLE

Regeneative the braking is a technology that converts the vehicle's kinetic energy (motion energy) into electrical energy during braking, Instead of wasting this energy as heat in the brake pads, the motor works as a generator, feeding the recovered energy back to the battery.

Key Points:

- Uses the electric motor as a generator during braking
- Stores energy in the battery or supercapacitor
- Improves overall vehicle efficiency

2. SYSTEM ARCHITECTURE

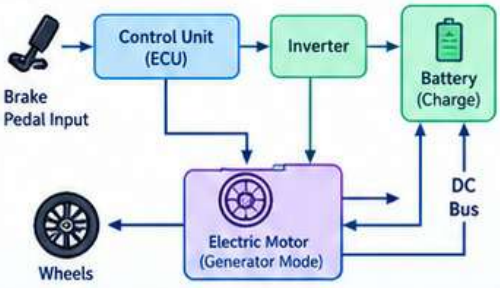


Figure: Block Diagram of Regenerative Braking System

3. HARDWARE / SOFTWARE USED

Hardware Components	Software / Tools
Electric Motor / Generator	MATLAB / Simulink
Power Inverter	Arduino IDE / Embedded C
Battery / Supercapacitor	Motor Control Library
Brake Control Unit (ECU)	CAN Communication
Sensors (Speed, Current, Temp)	Simulation Tools

4. METHODOLOGY

1	Sense	Detect vehicle speed, brake pressure, motor position, battery status.
2	Control	ECU decides the optimal regenerative torque using control algorithms.
3	Generate	Motor works as a generator and converts kinetic energy to electrical energy.
4	Store	Energy is stored in the battery/ supercapacitor.
5	Manage	Balance with normal braking and ensure safety & efficiency.

5. DETAILED CIRCUIT / BLOCK DIAGRAM

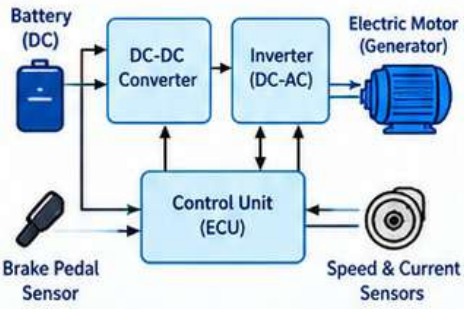
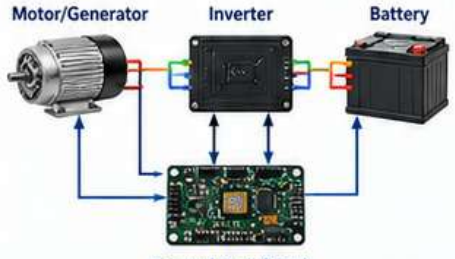


Figure: Regenerative Braking Circuit Diagram

6. FLOW / OPERATION

Brake Applied
(Driver presses brake pedal)
↓
ECU Calculates
(Braking force & motor torque)
↓
Motor Acts as Generator
(Converts kinetic → electrical energy)
↓
Energy Stored
(Battery / Supercapacitor)
↓
Vehicle Slows Down
(Controlled braking)

7. EXPERIMENTAL SETUP



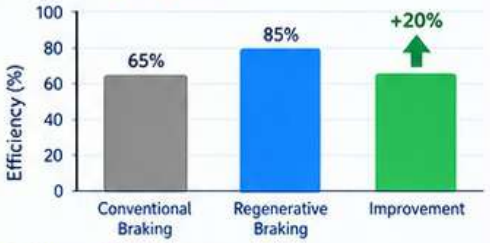
Actual hardware setup for regenerative braking.

8. RESULTS & OUTPUTS

A. Energy Recovery (Sample Data)

Parameter	Value
Initial Speed	60 km/h
Braking Energy Recovered	~ 15 – 30%
Battery Charge Increase	5 – 10%
Range Extension	5 – 15%

B. Efficiency Comparison



C. Energy Flow Diagram

Kinetic Energy → Generator (Motor) → Electrical Energy → Battery / Storage

9. PERFORMANCE ANALYSIS

Parameter	Value / Improvement
Energy Recovery Efficiency	15 – 30%
Overall Efficiency	85 – 95%
Battery Life	+20 – 40%
Brake Pad Wear	Reduced by 30 – 60%
CO_2 Emissions	Reduced

10. CONCLUSION & FUTURE SCOPE

Conclusion:

- Regenerative braking system recovers wasted energy, improves efficiency, extends battery life and reduces emissions.
- It is a key technology for sustainable and electric mobility.

Future Scope:

- Integration with smart grid (vehicle-to-grid).
- Use of supercapacitors for faster charge/discharge.
- Advanced control algorithms using AI/ML.
- Application in heavy vehicles and public transport.

11. REFERENCES

1. Electric Vehicle Technology – IET (https://www.theiet.org)
2. Regenerative Braking – SAE International (https://www.sae.org)
3. MATLAB & Simulink Documentation (https://in.mathworks.com)
4. IEEE Transactions on Vehicular Technology